

Quantum Information Theory

Operational Distance for Quantum States: Trace Distance & Helstrom Discrimination

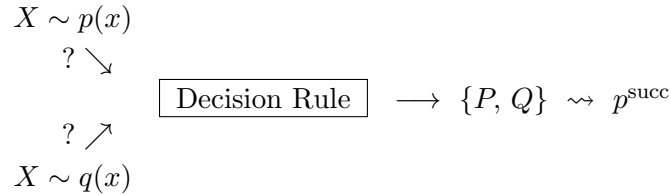
Talk 2 — June 25, 2026

Main idea: introduce a distance measure between two quantum states and relate it to something physically measurable.

1 Classical Case

We are given a sample realization x (with $x \in \mathbb{R}$, say) that could be drawn from one of two probability distributions, $p(x)$ or $q(x)$, with probability $\frac{1}{2}$ each.

Given $\{x, p(x), q(x)\}$ we want to make the best choice / guess of which pdf the sample x came from. We want a *decision rule*, optimal in some sense, that chooses $X \sim p(x)$ or $X \sim q(x)$ with the highest probability of success, p^{succ} .



1.1 Systematically designing an optimal decision rule

Let d be a random variable taking values in $\{P, Q\}$, representing the two distributions $p(x), q(x)$ (a label). We observe a random variable X drawn from one of the two distributions:

$$p(X | d = P) = p(X), \quad p(X | d = Q) = q(X).$$

We have the (uniform) prior

$$P(d = P) = P(d = Q) = \frac{1}{2}.$$

Using Bayes' law,

$$p(d, X) = p(d | X) p(X) = p(X | d) p(d),$$

so that

$$p(d | X) = \frac{p(X | d) p(d)}{p(X)}.$$

Given an observation X , choose the d that maximizes $p(d | X)$. On the right-hand side:

- $p(d) = \frac{1}{2}$ is a constant, independent of d ;

- $p(X)$ is independent of d .

Therefore

$$p(d \mid X) \propto p(X \mid d).$$

Take the optimal d given observed X to be the value $d = d^*$ that maximizes $p(d \mid X)$:

$$d^* = \arg \max_d p(d \mid X) \quad (\text{not known directly}) = \arg \max_d p(X \mid d) \quad (\text{known directly}),$$

since $p(X \mid d = P) = p(x)$ and $p(X \mid d = Q) = q(x)$.

1.2 The decision rule

$$\text{Given } x, \text{ choose } \begin{cases} P & \text{if } p(x) > q(x), \\ Q & \text{if } q(x) > p(x). \end{cases}$$

Performance of the decision rule. What is p^{succ} for given $p(x), q(x)$? Generally, computing the performance is more involved than deriving the optimal decision rule. Write

$$p^{\text{succ}} = P(d^* = d_c),$$

where $d_c \in \{P, Q\}$ is the chosen source distribution of X , and we want the marginal over P and Q .

Write the full joint distribution of all variables, $P(d^*, x, d_c)$. Since d^* is a function of x and we are interested in the event $d^* = d_c$,

$$p^{\text{succ}} = P(d^* = d_c) = \sum_{\substack{\forall x \\ d_c \in \{P, Q\}}} P(d^*(x) = d_c, x, d_c).$$

Using the Markov decomposition $P(A, B, C) = P(A \mid B, C) P(B \mid C) P(C)$,

$$p^{\text{succ}} = \sum_{x, d_c} P(d^*(x) = d_c \mid x, d_c) \underbrace{P(x \mid d_c)}_{\substack{p(x) \text{ for } d_c=P \\ q(x) \text{ for } d_c=Q}} \underbrace{P(d_c)}_{\frac{1}{2}}.$$

But we can write

$$P(d^*(x) = d_c \mid x, d_c) = \begin{cases} \mathbb{I}[p(x) > q(x)] & \text{when } d_c = P, \\ \mathbb{I}[q(x) > p(x)] & \text{when } d_c = Q, \end{cases}$$

where $\mathbb{I}[\cdot]$ is the indicator function. So

$$p^{\text{succ}} = \sum_{x: p(x) > q(x)} 1 \cdot p(x) \cdot \frac{1}{2} + \sum_{x: q(x) > p(x)} 1 \cdot q(x) \cdot \frac{1}{2} = \frac{1}{2} \left(\sum_{x: p > q} p(x) + \sum_{x: q > p} q(x) \right),$$

which gives the compact form

$$p^{\text{succ}} = \frac{1}{2} \sum_x \max(p(x), q(x)).$$

Using the identity $\max(a, b) = \frac{a + b + |a - b|}{2}$, and noting $\sum_x p(x) = \sum_x q(x) = 1$,

$$p^{\text{succ}} = \frac{1}{4} \sum_x (p(x) + q(x) + |p(x) - q(x)|) = \frac{1}{2} \left(1 + \frac{1}{2} \sum_x |p(x) - q(x)| \right).$$

2 Total Variation

Define the *total variation* distance as

$$TV(p, q) = \frac{1}{2} \sum_x |p(x) - q(x)| = \frac{1}{2} \|p(x) - q(x)\|_1,$$

where $\|f\|_1 = \sum_x |f(x)| = \int |f(x)| dx$ is the one-norm. Then

$$p^{\text{succ}} = \frac{1}{2} \left(1 + \frac{1}{2} TV(p, q) \right).$$

Properties of $TV(p, q) = \frac{1}{2} \|p - q\|_1$

- 1. Minimum value 0.** For each x in the sum, the term is minimized by having $p(x) = q(x)$. Then the distributions are indistinguishable, and

$$p^{\text{succ}} = \frac{1}{2} \left(1 + \frac{1}{2} \cdot 0 \right) = \frac{1}{2} \rightarrow \text{random choice.}$$

- 2. Maximum value 1.** Occurs when the supports of $p(x)$ and $q(x)$ are disjoint, where $\text{supp } f(x) = \{x \mid f(x) \neq 0\}$. Suppose p, q are disjoint:

$$\begin{aligned} \frac{1}{2} \sum_x |p(x) - q(x)| &= \frac{1}{2} \left(\sum_{\substack{x \\ p \neq 0, q=0}} |p(x) - q(x)| + \sum_{\substack{x \\ p=0, q \neq 0}} |p(x) - q(x)| \right) \\ &= \frac{1}{2} \sum_{\substack{x \\ p \neq 0, q=0}} p(x) + \frac{1}{2} \sum_{\substack{x \\ p=0, q \neq 0}} q(x) \quad (\text{disjoint}) \\ &= \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$

(Overlap always decreases the sum.)

- 3. $TV(p, q) = \frac{1}{2} \|p - q\|_1$ is a true metric distance,** so the pdf's form a metric space:

- $TV(p, q) \geq 0$;
- $TV(p, q) = TV(q, p)$;
- $TV(p, q) = 0 \iff p = q$;
- triangle inequality.

(One can do limits, completion, etc. as usual in abstract metric spaces.)

3 Quantum Binary Discrimination

Same idea as classical discrimination, but between two quantum states: density operators / matrices ρ and σ .

3.1 Review

An observable is a Hermitian operator / matrix A . It admits an eigendecomposition:

$$A = \sum_a a |\psi_a\rangle\langle\psi_a| = \sum_a a P_a,$$

where the a are real eigenvalues and measured physical outcomes (e.g. energy, position), the $|\psi_a\rangle$ are orthonormal eigenstates, and P_a is a projection operator. Sometimes written $\sum_a a |a\rangle\langle a|$, with a now a label (a real number).

An ensemble of quantum states, each with an assigned probability $\{p_i, |\psi_i\rangle\}$, is represented as a density operator / matrix:

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i| \longrightarrow \text{a quantum version of a classical pd function.}$$

Given a quantum state ρ and an observable A , the probability of measuring an outcome a is

$$P(a | \rho) = \text{Tr } P_a \rho.$$

Why? Because

$$\begin{aligned} \text{Tr } P_a \rho &= \text{Tr} \left(|a\rangle\langle a| \cdot \sum_i p_i |\psi_i\rangle\langle\psi_i| \right) = \text{Tr} \sum_i p_i \langle a | \psi_i \rangle \langle \psi_i | a \rangle \\ &= \sum_i p_i |\langle a | \psi_i \rangle|^2 = \sum_i p(\psi_i) p(a | \psi_i), \end{aligned}$$

so that

$\text{Tr } P_a \rho = p(a).$

So projections like P_a describe these kinds of measurement outcomes.

Properties of the projectors.

- (1) $P_a \geq 0$ (positive semi-definite);
- (2) $P_a^2 = P_a$, $P_a P_b = P_a \delta_{ab}$ (projections + orthogonality);
- (3) $\sum_a P_a = I$ (completeness).

3.2 POVMs

In more general cases — coarse graining, noisy measurements, or attaching an ancilla system, performing a projective measurement, and discarding — we use more general operators: POVMs, E_k .

Properties.

- (1) Born outcome probability rule: $P(k | \rho) = \text{Tr } E_k \rho$;
- (2) $E_k \geq 0$;
- (3) $\sum_k E_k = I$.

(Properties 1–3 are the same as for PVM projective measurements.) The following differ from PVMs:

- (4) Not orthogonal: $E_\ell E_m \neq \delta_{\ell m} E_\ell$ in general;
- (5) Not projections: $E_k^2 \neq E_k$ in general.

4 Two-Outcome POVM Discrimination

We are given two possible states ρ_0, ρ_1 . We will be handed one to use, with probability p_0 for ρ_0 and p_1 for ρ_1 , where $p_0 + p_1 = 1$. We perform a two-outcome measurement to decide which state, using a POVM $\{E_0, E_1\}$. Let $p_0 = p$ and $p_1 = 1 - p$.

1. We have $E_0 + E_1 = I$ and $E_0, E_1 \geq 0$. So the POVM is $\{E_0, I - E_0\}$; call $E_0 = E$. Then $I - E \geq 0 \Rightarrow E \leq I$, so

$$0 \leq E \leq I \quad \longrightarrow \quad \text{all eigenvalues of } E \text{ satisfy } 0 \leq \lambda \leq 1.$$

2. We have two quantum states prepared: ρ_0 with $P(\rho_0) = p$, and ρ_1 with $P(\rho_1) = 1 - p$. (Compare the classical case of pdf's $\{p, q\}$ with probability $\frac{1}{2}$ each.)

3. Perform a measurement:

$$\begin{aligned} \text{Outcome } E_0 = E &\Rightarrow \text{decide } \rho_0 \text{ is the quantum state,} \\ \text{Outcome } E_1 = I - E &\Rightarrow \text{decide } \rho_1 \text{ is the quantum state.} \end{aligned}$$

4. We have two ways of deciding correctly:

$$\begin{aligned} \text{True state } \rho_0 \text{ (prob. } p) : & \quad \text{Prob. of outcome } E_0 = E \text{ is } \text{Tr } E \rho_0; \\ \text{True state } \rho_1 \text{ (prob. } 1 - p) : & \quad \text{Prob. of outcome } E_1 = I - E \text{ is } \text{Tr}(I - E) \rho_1. \end{aligned}$$

Total:

$$P^{\text{corr}} = p \text{Tr } E \rho_0 + (1 - p) \text{Tr}(I - E) \rho_1.$$

We want to choose an E , given $\{p, \rho_0, \rho_1\}$, that maximizes P^{corr} — an optimization problem.

5. Now expand and rearrange; we get something similar to the classical case:

$$\begin{aligned} P^{\text{corr}} &= p \text{Tr } E \rho_0 + (1 - p) \text{Tr}(I - E) \rho_1 \\ &= p \text{Tr } E \rho_0 + (1 - p) [\text{Tr } \rho_1 - \text{Tr } E \rho_1] \\ &= p \text{Tr } E \rho_0 + \text{Tr } \rho_1 - \text{Tr } E \rho_1 - p \text{Tr } \rho_1 + p \text{Tr } E \rho_1 \\ &= p \text{Tr } E \rho_0 + (1 - p) - (1 - p) \text{Tr } E \rho_1 \\ &= (1 - p) + \text{Tr } E(p \rho_0 - (1 - p) \rho_1). \end{aligned}$$

So

$$P^{\text{corr}} = (1 - p) + \text{Tr } E(p \rho_0 - p_1 \rho_1) \xrightarrow{p=1/2} \frac{1}{2} + \frac{1}{2} \text{Tr } E(\rho_0 - \rho_1),$$

compare with the classical $\frac{1}{2} + \frac{1}{2} \sum_x |p(x) - q(x)| = \frac{1}{2} + \frac{1}{2} \|p - q\|_1$.

Define the *Helstrom operator*

$$\Delta = p_0 \rho_0 - (1 - p) \rho_1 = p_0 \rho_0 - p_1 \rho_1,$$

which encodes the *known* problem information. So

$$P^{\text{corr}} = 1 - p + \text{Tr } E \Delta \quad \longrightarrow \quad \text{for the given } \Delta, \text{ adjust } E \text{ with } 0 \leq E \leq I \text{ to maximize } P^{\text{corr}}.$$

Note we are optimizing over an operator (matrix) in an expression involving a trace, under the psd constraints $0 \leq E \leq I$.

5 The Optimization Problem

For given $\Delta = p_0\rho_0 - p_1\rho_1$, find

$$E^* = \arg \max_{0 \leq E \leq I} \text{Tr } E\Delta,$$

where $0 \leq E \leq I$ means all eigenvalues of E lie in $[0, 1]$.

We know ρ_0, ρ_1 are psd (all eigenvalues $\lambda \geq 0$) and $\text{Tr } \rho_0 = \text{Tr } \rho_1 = 1$. But the weighted difference $p_0\rho_0 - p_1\rho_1$ is still Hermitian, yet not necessarily psd (some eigenvalues < 0 are possible).

Suppose we set $E = I$. Then

$$\text{Tr } I\Delta = \text{Tr } \Delta = \sum_i \delta_i \quad (\delta_i = \text{eigenvalues of } \Delta).$$

Generally some $\delta_i > 0$ and some $\delta_i < 0$, so the sum will be at least partially decreased by the negative δ_i .

Main idea: choose E as a *projector onto only the positive eigenvalue subspace* of Δ . Then $\text{Tr } E\Delta$ will sum only positive values. Design E to maximize this.

5.1 The optimal E

Claim. Eigendecompose Δ :

$$\Delta = \sum_i \delta_i |i\rangle\langle i| = \sum_j \delta_j^+ |j^+\rangle\langle j^+| + \sum_k \delta_k^- |k^-\rangle\langle k^-|,$$

where $\{\delta_j^+\}$ are the positive eigenvalues and $\{\delta_k^-\}$ are the zero or negative eigenvalues. So $\Delta = \Delta_+ + \Delta_-$ with

$$\Delta_+ = \sum_j \delta_j^+ |j^+\rangle\langle j^+|, \quad \Delta_- = \sum_k \delta_k^- |k^-\rangle\langle k^-|.$$

Then

$$E^* = \sum_j |j^+\rangle\langle j^+| \equiv \Pi_+,$$

a projection onto the positive eigenspace of Δ .

Why.

1. *Notation.*

- (i) $A > 0$ means, for all unit v , $\langle v|A|v\rangle > 0 \quad \forall v$.
- (ii) $A < I$ means, for all v with $|v| = 1$,

$$A - I < 0 \implies \langle v|A - I|v\rangle < 0 \implies \langle v|A|v\rangle - \underbrace{\langle v|v\rangle}_1 < 0 \implies \langle v|A|v\rangle < 1 \quad \forall v.$$

2. *Upper bound.* We have $\Delta = \sum_i \delta_i |i\rangle\langle i|$, so

$$\text{Tr } E\Delta = \text{Tr } E \left(\sum_i \delta_i |i\rangle\langle i| \right) = \sum_i \delta_i \text{Tr } (E |i\rangle\langle i|) = \sum_i \delta_i \langle i|E|i\rangle.$$

We know (i) $0 \leq \langle i|E|i \rangle \leq 1$ and (ii) some $\delta_i > 0$, some $\delta_i \leq 0$. We can maximize this sum term by term by setting

$$\langle i|E|i \rangle = 1 \text{ for } \delta_i > 0, \quad \langle i|E|i \rangle = 0 \text{ for } \delta_i \leq 0.$$

(We don't know yet that this is achievable, but it does set an upper bound.) Therefore

$$\boxed{\text{Tr } E\Delta \leq \sum_i \delta_i^+ = \text{Tr } \Delta_+ .}$$

3. Achievability. This bound is achieved by setting

$$\boxed{E^* = \sum_j |j^+\rangle \langle j^+| \equiv \Pi_+ .}$$

Why? Because

$$\begin{aligned} \text{Tr } E^* \Delta &= \text{Tr} \left(\sum_j |j^+\rangle \langle j^+| \cdot \sum_i \delta_i |i\rangle \langle i| \right) = \text{Tr} \sum_{j,i} \delta_i |j^+\rangle \underbrace{\langle j^+|i\rangle}_{\delta_{j+i}} \langle i| \\ &= \sum_i \delta_i^+ = \text{Tr } \Delta_+ . \end{aligned}$$

Hence the optimal POVM element is

$$\boxed{E^* = \sum_i |i^+\rangle \langle i^+| .}$$

6 The Probability of Success

Now look at p^{succ} . For $\Delta = p_0\rho_0 - p_1\rho_1$,

$$p^{\text{succ}} = 1 - p + \text{Tr } E^* \Delta = 1 - p + \text{Tr } \Delta_+ .$$

Now

$$\text{Tr } \Delta_+ = \sum_{i: \delta_i > 0} \delta_i = \frac{1}{2} \left(\sum_i |\delta_i| + \sum_i \delta_i \right) = \frac{1}{2} (\|\Delta\|_1 + \text{Tr } \Delta),$$

and

$$\text{Tr } \Delta = \text{Tr} (p\rho_0 - (1-p)\rho_1) = p \text{Tr } \rho_0 - (1-p) \text{Tr } \rho_1 = p - (1-p) = 2p - 1,$$

so that

$$\text{Tr } \Delta_+ = \frac{1}{2} (\|\Delta\|_1 + 2p - 1).$$

Therefore

$$p^{\text{succ}} = 1 - p + \frac{1}{2} \|\Delta\|_1 + p - \frac{1}{2} = \frac{1}{2} (1 + \|\Delta\|_1).$$

So the optimal probability of success in binary discrimination is

$$\boxed{p^{\text{succ}} = \frac{1}{2} \left(1 + \|p\rho_0 + (1-p)\rho_1\|_1 \right) .}$$

Comparison to the classical $p = \frac{1}{2}$ case.

$$\boxed{p^{\text{succ}} = \frac{1}{2} \left(1 + \frac{1}{2} \|p - q\|_1 \right)} \quad (\text{classical}),$$

$$\boxed{p^{\text{succ}} = \frac{1}{2} \left(1 + \frac{1}{2} \|\rho_0 - \rho_1\|_1 \right)} \quad (\text{quantum}).$$

7 Quantum Trace Distance of States

(For $p = \frac{1}{2}$.) Define

$$T(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1, \quad \|A\|_1 \equiv \sum_i |\lambda_i|,$$

alongside the classical

$$TV(p, q) = \frac{1}{2} \|p - q\|_1, \quad \|f\|_1 \equiv \sum_x |f(x)|.$$

Same general properties.

1. Both are proper metric distances.
2. Minimum 0: $\rho = \sigma \iff T = 0 \rightarrow p^{\text{succ}} = \frac{1}{2}$.
3. Maximum 1: ρ, σ perfectly distinguishable with $p^{\text{succ}} = 1$.
4. Perfect distinguishability also has a disjointness requirement.

Perfect distinguishability. One can show that for $T(\rho, \sigma) = 1$:

- (1) for $\delta_i > 0$: $\langle i | \sigma | i \rangle = 0$,
- (2) for $\delta_i < 0$: $\langle i | \rho | i \rangle = 0$.

Or, equivalently:

- (1) all positive eigenvectors of $\rho - \sigma$ belong to the image of ρ ;
- (2) all negative eigenvectors of $\rho - \sigma$ belong to the image of σ .